Dashboard and Visualisation

COMP5339 Assignment 2 | Member D | A-generated data verified 31 October 2026

Data source and two validation scopes

A's supplied-generator output covers 414 eligible DC stations and 1,610 connectors, with session starts over one day from 1 October 2026. We checked the complete 172,803-event stream against C's stored history, and separately replayed A's unchanged 36-event session through real MQTT and D's browser into an empty event store. Original event IDs, timestamps and payloads were retained. These are generated simulations, not observed EV transactions.

Measure	Complete stored dataset	A-generated MQTT demo
Events	172,803	36
Price updates	7,038	2
Meter readings	155,597	32
Completed sessions	5,084	1
Completed energy (kWh)	129,422.20	22.60
Completed revenue (AUD)	88,908.69	16.72
Active sessions at the end	0	0

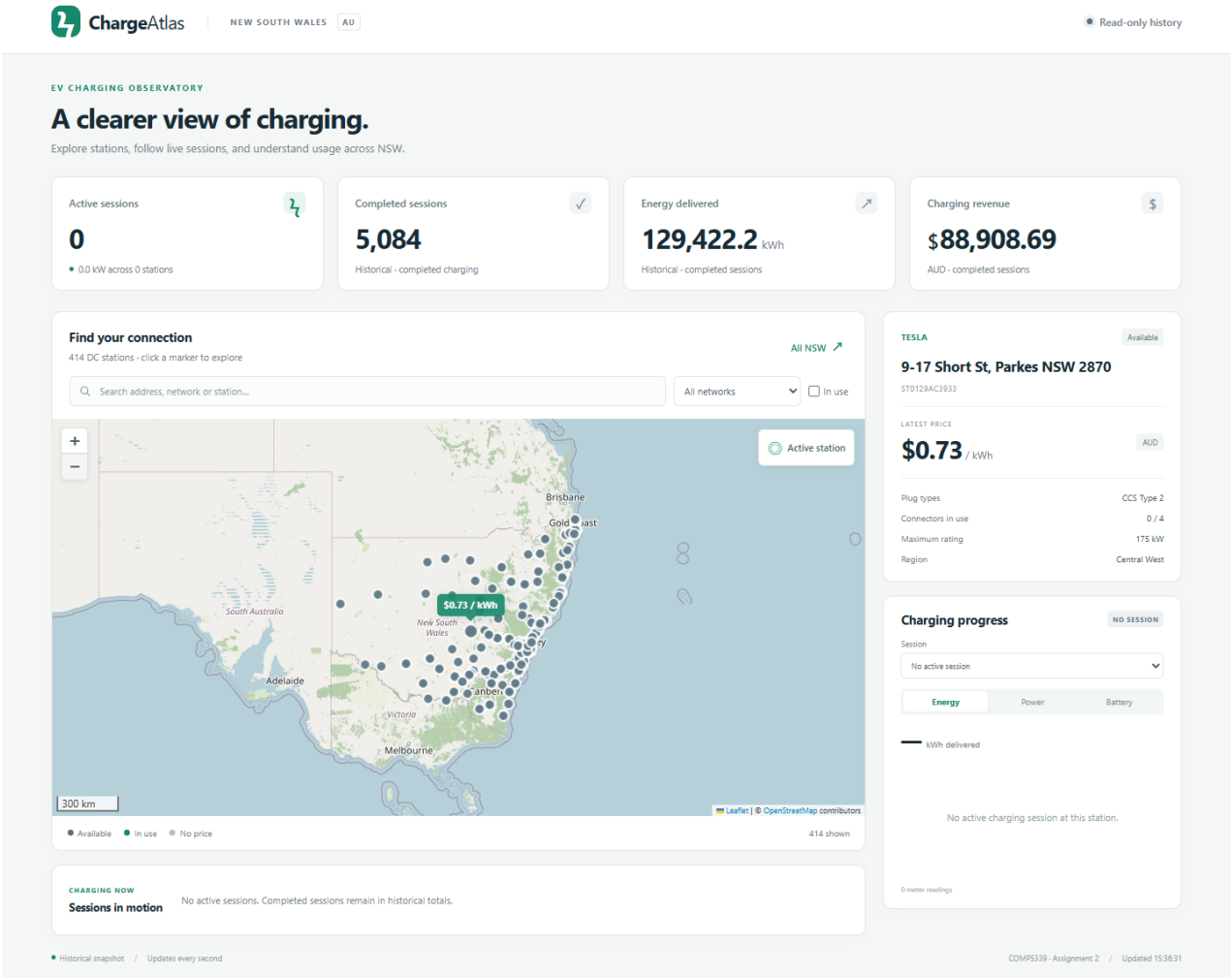


Figure D1. Read-only display of the complete stored dataset. Zero active sessions is expected after all generated sessions have ended. Full-stream storage was reconciled offline; this is not a full-stream MQTT replay.

Live integration and verified behaviour

Same application and data lineage | Independent, empty-event replay database

A single Flask runtime joins B's live state with C's repository by A1 station ID. The map, details and energy/power/SoC curves update every second. Historical curves remain queryable after sessions end. Missing source metadata is displayed explicitly.

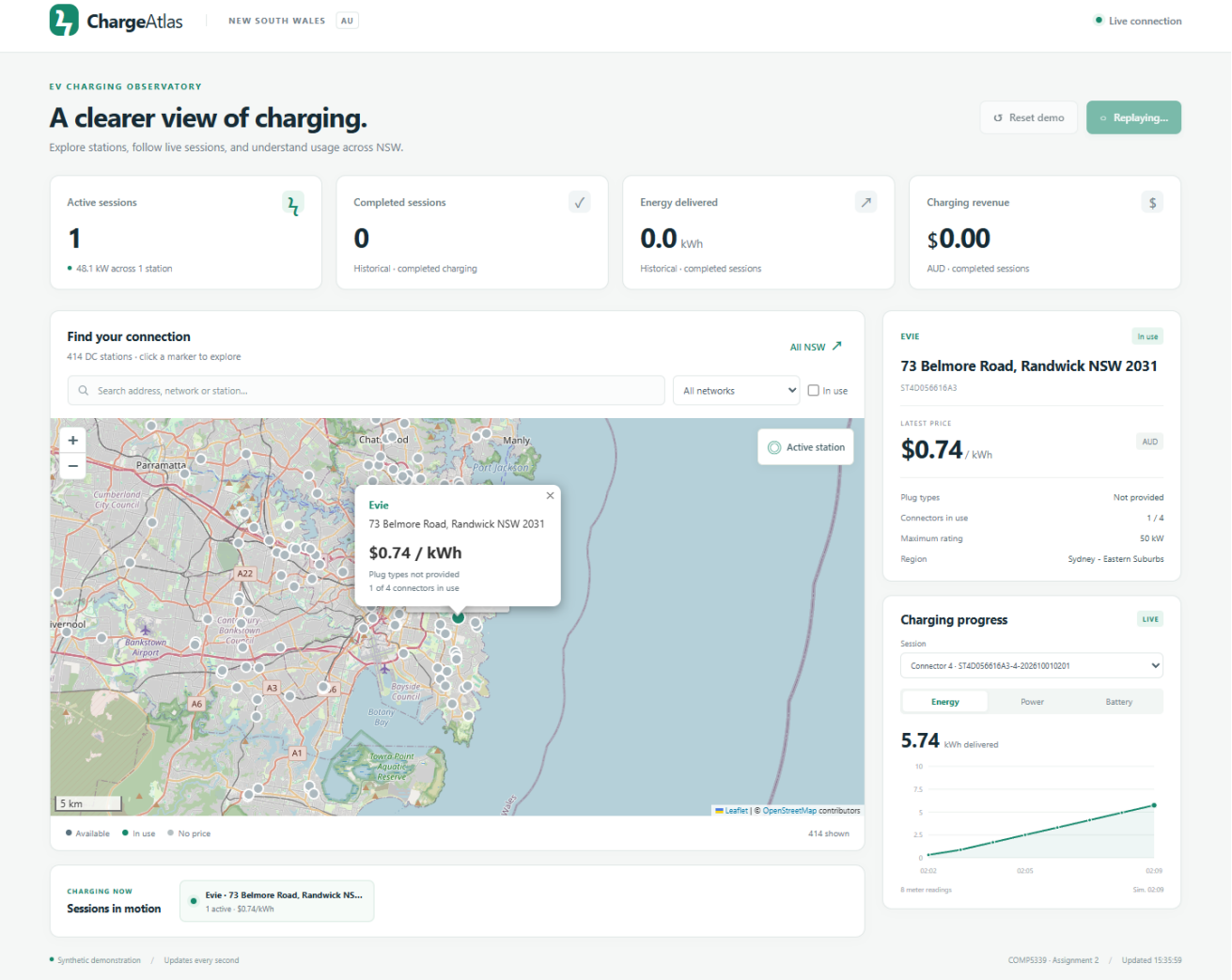


Figure D2. A's generated MG4 session at Evie, 73 Belmore Road, Randwick, is active. The curve grows while historical cards remain zero because they count completed sessions only.

Measured result and validation

All 36 messages were published with QoS 1 and one-second spacing (minimum gap 1.000 s). The station price changes from AUD 0.75 to 0.74/kWh before charging. At completion there is one completed session, zero active sessions, 22.60 kWh, AUD 16.72 and 32 curve points. All 52 regression tests passed; this additional browser run had no JavaScript page errors. Reset restored empty event history; source database hashes were unchanged.

Interpretation and limitations

The two result scopes are not combined. The complete dataset averages 30.61 charging minutes and peaks at 183.4 kW. Full-stream paced MQTT replay and production outage recovery were not tested. Upstream plug types are missing for 248 stations, including the replay site; base-map tiles need internet.

Reproduce: Start_A_Data_Demo.cmd, then Run demo. The earlier hand-crafted UI fixture is excluded from these reported results.